Lazar Paroski

+1 519-498-9505 | lazar@live.ca | linkedin.com/in/lazar-paroski | lazarparoski.com

Research Interests

Quantum photonics, cavity QED, and light-matter interfaces; emergent phenomena in hybridized light-matter systems; physics of 2D and topological materials.

Education

Queen's University Sept 2022 – April 2027

Bachelor of Applied Science, Major in Engineering Physics and Computing *Kingston, ON*

- Harold M. Cave Scholarship in Experimental Physics, 2024-2025
- Dean's List Scholar, 2025-2026, 2024-2025, 2023-2024, 2022-2023
- Cumulative GPA: 3.90 (Jan 2024 onward)

Simons SCEECS Summer School June 2025

Participant, Simons Collaboration on Extreme Electrodynamics of Compact Sources (SCEECS) *San Antonio, TX*

- Selected as the only Canadian student to attend the summer school held at the University of Texas, San Antonio.
- Completed intensive lectures and tutorials on extreme astrophysical and laboratory plasmas, including MHD, shocks, turbulence, reconnection, relativistic flows, and laser–plasma interactions.

Research Experience

USEQIP Participant & Undergraduate Research Award Recipient May 2026 – Aug 2026

Institute for Quantum Computing (IQC), University of Waterloo (Full-Time) *Waterloo, ON*

- Selected for the Undergraduate School on Experimental Quantum Information Processing (USEQIP), a competitive program offering intensive training in quantum information science alongside leading researchers.
- Awarded an Undergraduate Research Award to join Prof. Michael Reimer's group at IQC, working to enhance the performance of nanowire quantum dot single-photon sources for scalable quantum networking applications.
- Assisting with ongoing experimental work on entanglement swapping, a foundational primitive for quantum repeaters and long-distance quantum networks.

Undergraduate Researcher May 2025 – Present

Shastri Lab, Queen's University (Full-Time) *Kingston, ON*

- Designed and constructed a novel optoelectronic Ising machine utilizing cascaded TFLN Mach-Zehnder modulators, SOAs, and an RFSoC 4x2 FPGA to physically accelerate combinatorial optimization problems.
- Developed a custom Ising-machine based computational framework to simulate protein folding, validating solver logic and characterizing algorithmic performance.
- Developing a specialized Ising machine architecture to investigate protein folding dynamics and explore methods for recovering physical intermediate states.

Research Assistant Aug 2024 – Present

Queen's University Dept. of Physics (Part-Time) *Kingston, ON*

- Investigated non-equilibrium heating dynamics in quantum spin chains under the supervision of Prof. Ivan Iorsh, specifically analyzing how the Kibble-Zurek mechanism influences excitation rates during periodic phase transitions.
- Developed GPU-accelerated simulations in Python (using CUDA) to model system evolution, enabling high-resolution analysis of topological defect densities and spatial correlations.
- Co-authored a manuscript on driven quantum systems, utilizing numerical simulations to characterize the heating dynamics and thermalization properties of integrable models.

Research Assistant May 2024 – Aug 2024

Sudbury Neutrino Observatory Laboratory – SNOLAB (Full-Time) *Sudbury, ON*

- Designed and implemented a cooling system for the photomultiplier tube (PMT) testing setup as part of the nEXO collaboration at SNOLAB.
- Enhanced Python scripts for PMT data analysis to improve accuracy, automation, and efficiency.
- Documented work in a technical report and presented findings to the SNOLAB research group.

Publications & Manuscripts

L. Paroski and I. V. Iorsh, *Heating dynamics in a periodically driven transverse field Ising chain*, manuscript in preparation (2026).

L. Paroski, *Photonic Ising Machine Architectures for HP Protein Folding*, manuscript in preparation under the supervision of B. Shastri (2026).

Projects

Mach–Zehnder Modulator PCB Integration & Device Characterization (Shastri Lab) 2025 – Present

- Assembled custom PCB platforms for TFLN Mach-Zehnder modulators by integrating precision bias circuitry, SMA RF interfaces, and thermal management solutions.
- Set up and characterized MZM operation by capturing eye diagrams and recording data such as quality factor, eye height/width, and duty cycle distortion.

Thermodynamics of Neural Network Training (ENPH 472 Final Project) 2025

- Modeled neural network training as a thermodynamic phase transition, implementing a Nosé-Hoover thermostat to measure specific heat capacity (C_v).
- Detected a massive Schottky anomaly and calculated the entropy difference ($\Delta S \approx 0.7 k_B$ per parameter) driving the transition from overfitting to generalization.

Selected Coursework

Quantum Mechanics I & II, Statistical Mechanics, Thermodynamics, Solid State Physics, Advanced Classical Mechanics (Dynamical Systems), Electromagnetism, Mathematical Methods I & II, Computational Engineering Physics, Algorithms & Data Structures, Computer Architecture.

Additional Experience

Student Athlete Academic Mentor Sep 2023 – Present

Queen's University (Part-Time) *Kingston, ON*

- Support first-year student-athletes' transition to post-secondary academics through tutoring and mentorship.
- Hold weekly one-on-one check-ins to monitor academic performance and well-being.

Math and Physics Content Developer May 2023 – Aug 2023

Maplesoft (Full-Time) *Waterloo, ON*

- Developed 30+ math and physics interactive modules for Maple Learn and published a blog post to engage new users.
- Secured 450 student licenses from French institutions by developing targeted educational content.

Extracurricular

Queen's Varsity Water Polo Jan 2024 – Present

- Compete as a member of a high-performance varsity team while demonstrating discipline, teamwork, and resilience under pressure.
- Manage a rigorous academic workload while committing 15+ hours per week to training, travel, and competition.
- Academic All-Canadian, 2025-2026, 2024-2025, 2023-2024, 2022-2023.

Skills

- Programming: Python, C, C++, VHDL, Java, Bash, Git, Linux, Maple, LabVIEW
- Scientific Computing: NumPy, SciPy, Matplotlib, PyTorch, CUDA, Qiskit, PennyLane, Slurm, HPC/Cluster Computing
- Experimental Physics: Photonic Hardware (TFLN Mach–Zehnder Modulators, Lasers, Polarizers, Optical Amplifiers), FPGA Development (RFSoC 4x2, PYNQ), RF Electronics, PCB Integration, PMT Characterization
- Lab Equipment: High-Speed Oscilloscopes, Pattern Generators, Amplifiers, Power Supplies, Soldering, Remote Lab Automation
- Documentation: LaTeX, Technical Writing, Experiment Logs and Reporting
- Languages: Fluent in English and French